

Learner Guide

International Sales — what to do, why, and where you are in the process | ★★★★★☆

This guide shows one sound way to complete the scenario, the reasoning behind each step, and which stage of the data analysis process each step belongs to. The trickiest parts are reshaping the exchange-rate table and getting the currency conversion the right way round. Work the task yourself first, then check your method here. Screenshots are illustrative mock-ups; Excel/Power Query is shown.

The data analysis process — follow it in order:

Identify → Collect → Clean/Prepare → Analyse → Visualise → Interpret/Report

Each step is tagged with its stage. Most of the marks here are in Clean/Prepare — the reshape, the December filter and the currency join.

Scenario 1 — Data Gathering and Validation

Step 1 — Identify what you need

STAGE 1 · IDENTIFY

What you do: Confirm the requirement: three sales files (one per rep/country/currency), the exchange-rate file, and the goal of one combined set with all prices converted to euros. Note the known issues — the rate file is transposed with title rows, only December 2018 is needed, and the conversion must be matched by currency.

Why start here? Spotting that the rate table needs reshaping and filtering before you can join it means you plan the work instead of hitting a wall at the conversion step.

Step 2 — Bring each source in as a connection

STAGE 2 · COLLECT

What you do: Import all four files as **connection only**.

1. **Data** → **Get Data** → **From File** → the CSV or Excel file.
2. At the load prompt, choose **only Create Connection**.

Figure 1 — Choose **Only Create Connection** so each source stays as a query.

Why connection only? The rate file needs reshaping before it's usable; a connection-only query lets you clean it in Power Query without dumping the messy layout onto a sheet.

Step 3 — Combine the three sales files

STAGE 3 · CLEAN / PREPARE

What you do: Double-click a sales query to open it, add a **Representative** column to each (Jake / Liam / Nick), confirm the column names match, then **Append Queries** to stack the three into one sales table. Remove duplicates and check for missing values.

Why append, and why a Rep column? The three files have identical columns for different people, so they stack (append). Without a Representative column you can't tell whose sale is whose once combined — and Task 3 of Scenario 2 depends on comparing reps.

Step 4 — Reshape the exchange-rate file (unpivot, then filter)

STAGE 3 · CLEAN / PREPARE

What you do: Open the rate query, remove the title rows, promote the real header, then **unpivot** the currency columns so each row is one month-currency-rate. Finally filter to **December 2018**.

Transpose vs Unpivot — don't confuse them. This is the step learners most often get wrong. The rate table has months down the rows and currencies across the columns. You need to **unpivot** it (turn the currency columns into rows), not **transpose** it (which only rotates the grid and leaves it just as wide). The figure below shows the difference.

Transpose vs Unpivot — they look similar but do different things

Start: months spread across columns (wide)

Currency	Jan	Feb	Mar
GBP	0.88	0.88	0.89

TRANSPOSE — flips the grid (still wide)

Currency	GBP
Jan	0.88
Feb	0.88
Mar	0.89

Headers and side column just swapped.

Still wide — can't merge on it. Rarely what you want.

UNPIVOT — columns become rows (long/tidy)

Currency	Month	Rate
GBP	Jan	0.88
GBP	Feb	0.88
GBP	Mar	0.89

Each month becomes its own row — one value per row. This is the shape a **merge or chart needs. Use this.**

Figure 2 — Transpose only rotates the grid (still wide); unpivot turns columns into rows (long/tidy), which is what a join needs.

Why unpivot, not transpose? A join needs one rate per row, keyed by currency. Transposing the table just swaps its orientation — it stays "wide" and still can't be joined. Unpivoting reshapes it into the tidy, long form the merge expects. As a rule: if you'll merge or chart it afterwards, you want unpivot.

Step 5 — Join the rates, matched by currency

STAGE 3 · CLEAN / PREPARE

What you do: Merge the December-2018 rates onto the combined sales table using **Currency** as the key, so each sale gains its correct rate.

After unpivoting, filter the rate table to December 2018, then join on Currency

Rep	Product	Unit Price (local)	Currency	Rate (Dec-18)
Liam	Tablet	810	GBP	0.897525
Nick	Computer	5940	CNY	7.839751
Jake	Computer	1100	EUR	1.000000

Only the December 2018 rates are needed — filter the month before joining.

Currency is the key. EUR = 1 (already in euros). BRL and MAD rates exist but no rep uses them — ignore.

⚠ Watch the direction: Unit Price EUR = local price / rate (e.g. 810 GBP / 0.8975 = €902.48).

Figure 3 — After unpivoting and filtering to December 2018, join the rate onto each sale by currency.

Why match by currency? Each rep sells in a different currency, so the right rate has to follow the currency, not the rep. Note that BRL and MAD rates exist in the file but no rep uses them — they're distractors and should simply not match. EUR's rate is 1 (Jake's prices are already in euros). Save as `Combined_Sales_2018.xlsx`.

Scenario 2 — Data Analysis

Step 6 — Convert unit price to euros

STAGE 4 · ANALYSE

What you do: Add a **Unit Price EUR** column, dividing the local price by the matched rate, formatted to 2 decimal places:

```
=[@[Unit Price Local Currency]] / [@[Exchange Rate]]
```

Why divide, and why watch the direction? The rates are quoted as "units of local currency per euro", so dividing converts local back to euros (e.g. 810 GBP ÷ 0.8975 = €902.48). Multiplying instead would inflate the figures massively — getting the direction right is exactly what's being tested.

Step 7 — Calculate revenue

STAGE 4 · ANALYSE

What you do: Add a **Revenue** column = Unit Price EUR × Quantity. Check for null Quantity or Unit Price first.

```
=[@[Unit Price EUR]] * [@Quantity]
```

Step 8 — Analyse performance

STAGE 4 · ANALYSE

What you do: Use a PivotTable to total revenue by representative and by product, and average revenue per product.

Expected results:

Best representative: Jake (France) with about **€1,155,100** — well ahead of Liam (UK, €315,969) and Nick (China, €276,139).

Top product by revenue: Computer (~**€665,586**), just ahead of Tablet (~€605,822).

Average revenue per unit: Computer €1,033.52, Tablet €840.25, Smartphone €554.06, Screen €232.86.

Step 9 — Visualise

STAGE 5 · VISUALISE

What you do: Chart total revenue by representative (and optionally by product) as a bar chart.

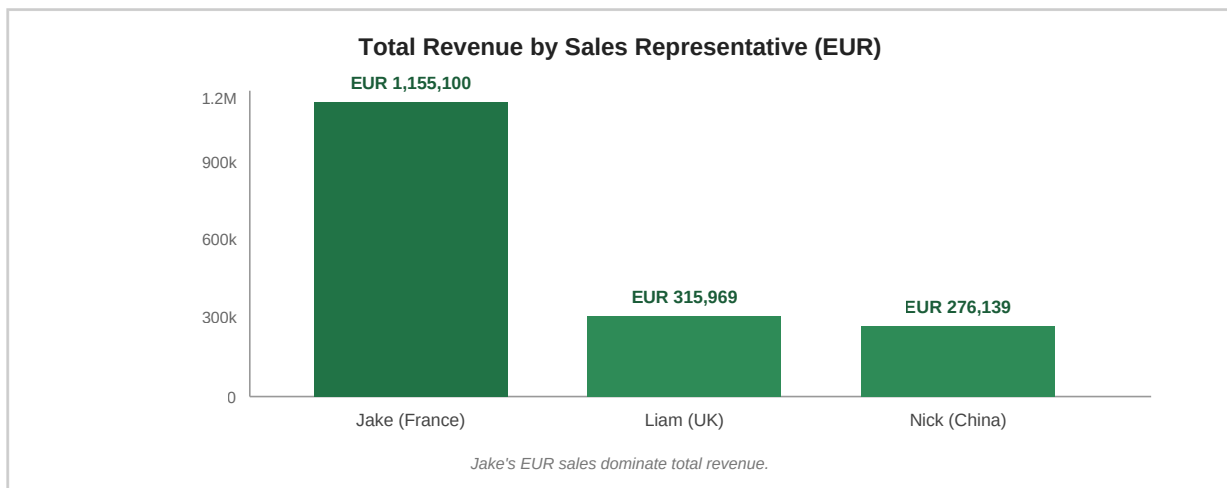


Figure 4 — Total revenue by representative; Jake's euro sales dominate.

Why chart it? The gap between Jake and the others is far clearer as bars than as a table of numbers — the visual makes the headline finding land immediately for a senior audience.

Step 10 — Interpret and report

STAGE 6 · INTERPRET / REPORT

What you do: Summarise for senior management beneath your charts, then save the workbook with its tables and charts intact.

"Once all sales were converted to euros, Jake (France) was the clear top performer at about €1.16m — roughly four times Liam or Nick. Computers generated the most revenue overall (~€666k), narrowly ahead of tablets, and also had the highest average revenue per unit (~€1,034). This points to France and the computer line as the strongest contributors, worth prioritising in planning."

Marker's checklist — process followed in order

- **Identify:** sources, the reshape/filter need and the conversion rule confirmed first.
- **Collect:** all four files brought in as **connection only** (not opened).
- **Clean/Prepare:** three sales files appended with a Representative column; rate table **unpivoted** (not transposed) and filtered to December 2018; rates joined by Currency.
- **Analyse:** Unit Price EUR by formula ($\div$ rate); Revenue = price $\times$ quantity; totals by rep and product.
- **Visualise:** revenue bar chart.
- **Interpret/Report:** findings summarised; saved as .xlsx.